Hyperbolas

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Standard form**

For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, find:

a the vertices $(\pm 4, 0)$.

b the foci $c^2 = 16 + 9 = 25$, so foci $(\pm 5, 0)$.

c the equations of the asymptotes $y = \pm\frac{3}{4}x$.

(a) $(\pm 4, 0)$.; (b) $c^2 = 16 + 9 = 25$, so foci $(\pm 5, 0)$.; (c) $y = \pm\frac{3}{4}x$.

2 **[Foundational]** • **Target: Standard form**

Write the standard-form equation of the hyperbola with vertices $(0, \pm 2)$ and foci $(0, \pm 3)$.

$a = 2$, $c = 3$, $b^2 = 9 - 4 = 5$: $\frac{y^2}{4} - \frac{x^2}{5} = 1$.

3 **[Foundational]** • **Target: Standard form**

Consider the equation $4x^2 - y^2 + 8x + 4y - 4 = 0$.

a Write the equation in standard form by completing the square. $4(x+1)^2 - (y-2)^2 = 4$, i.e. $(x+1)^2 - \frac{(y-2)^2}{4} = 1$.

b Identify the conic and state its center. A hyperbola opening left-right with center $(-1, 2)$.

(a) $4(x+1)^2 - (y-2)^2 = 4$, i.e. $(x+1)^2 - \frac{(y-2)^2}{4} = 1$.; (b) A hyperbola opening left-right with center $(-1, 2)$.

4 **[Foundational]** • **Target: Identifying conics**

Identify each conic section from its equation:

a $x^2 + y^2 = 49$ Circle.

b $\frac{x^2}{9} + \frac{y^2}{25} = 1$ Ellipse.

c $y^2 - 4x^2 = 16$ Hyperbola.

d $y = 3x^2 - 6x + 1$ Parabola.

(a) Circle.; (b) Ellipse.; (c) Hyperbola.; (d) Parabola.

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5 [Foundational] • Target: Identifying conics

A hyperbola has asymptotes $y = \pm 2x$ and vertices $(\pm 3, 0)$. Find its equation.

$a = 3$ and $\frac{b}{a} = 2$ gives $b = 6$: $\frac{x^2}{9} - \frac{y^2}{36} = 1$.

6 [Foundational] • Target: Hyperbolas centered away from the origin

For the hyperbola $\frac{(y+1)^2}{4} - \frac{(x-3)^2}{9} = 1$, find:

a the center $(3, -1)$

b the vertices $(3, -1 \pm 2) = (3, 1), (3, -3)$

c the direction of opening opens up/down

(a) $(3, -1)$; (b) $(3, -1 \pm 2) = (3, 1), (3, -3)$; (c) opens up/down

7 [Foundational] • Target: Hyperbolas centered away from the origin

Complete the square to write $9x^2 - 4y^2 - 18x - 16y - 43 = 0$ in standard form, and state the center.

$9(x^2 - 2x) - 4(y^2 + 4y) = 43$: $9(x-1)^2 - 9 - 4(y+2)^2 + 16 = 43$, so $9(x-1)^2 - 4(y+2)^2 = 36$, i.e. $\frac{(x-1)^2}{4} - \frac{(y+2)^2}{9} = 1$. Center: $(1, -2)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Eccentricity and applications

Find the eccentricity of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$, to two decimal places, and state whether it is closer to a pair of straight lines or to a circle.

$c = \sqrt{9 + 16} = 5$: $e = \frac{5}{3} \approx 1.67$. Since $e > 1$ and relatively large, the branches are wide and open quickly, resembling straight lines more than a tight curve.

9 [Intermediate] • Target: Eccentricity and applications

A ship uses two radio towers 200 km apart (foci) to navigate via hyperbolic positioning: the difference in distances from the ship to each tower is constant at 120 km. Write the equation of the hyperbola (centered at the midpoint of the towers, opening left-right), in standard form.

$2a = 120$, so $a = 60$; $2c = 200$, so $c = 100$; $b^2 = c^2 - a^2 = 10000 - 3600 = 6400$. Equation: $\frac{x^2}{3600} - \frac{y^2}{6400} = 1$.

10 [Intermediate] • Target: Graphing a hyperbola branch

The upper-right branch of $y^2 - x^2 = 4$ (as $y = \sqrt{4 + x^2}$) is shown. Use the graph to estimate y at $x = 3$, then verify algebraically, and compare to the asymptote $y = x$.

From the graph, $y \approx 3.6$ at $x = 3$. Algebraically: $y = \sqrt{4 + 9} = \sqrt{13} \approx 3.61$, which is close to but above the asymptote value $y = x = 3$, consistent with the branch approaching but never touching its asymptote.

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11 **[Intermediate]** • **Target: The hyperbola's defining property**

State the geometric (difference-of-distances) definition of a hyperbola, and use it to verify the vertex $(4, 0)$ of $\frac{x^2}{16} - \frac{y^2}{9} = 1$ (foci $(\pm 5, 0)$) satisfies it.

A hyperbola is the set of points where the difference of distances to the two foci is constant ($= 2a$). At $(4, 0)$: distance to $(5, 0)$ is 1; distance to $(-5, 0)$ is 9. Difference $= 9 - 1 = 8 = 2a = 2(4)$ ✓.

12 **[Intermediate]** • **Target: Comparing all four conic sections**

State the key algebraic feature (coefficient signs) that distinguishes a circle, ellipse, parabola, and hyperbola in a general second-degree equation $Ax^2 + By^2 + \ldots = 0$.

Circle: $A = B$ (same sign, equal). Ellipse: A, B same sign, unequal. Parabola: exactly one of A, B is zero (only one variable squared). Hyperbola: A, B opposite signs.

13 **[Intermediate]** • **Target: Finding the equation from asymptotes and a point**

A hyperbola centered at the origin has asymptotes $y = \pm\frac{2}{3}x$ and passes through $(3, 0)$. Find its equation.

Since it passes through $(3, 0)$, it opens left-right with $a = 3$. From $\frac{b}{a} = \frac{2}{3}$: $b = 2$. Equation: $\frac{x^2}{9} - \frac{y^2}{4} = 1$.

14 **[Intermediate]** • **Target: Applications: LORAN navigation**

Two LORAN radio towers are 260 km apart. A ship determines the difference in distances to the two towers is 100 km. Write the equation of the hyperbola of possible ship locations (centered at the midpoint, opening along the line connecting the towers).

$2a = 100 \Rightarrow a = 50$; $2c = 260 \Rightarrow c = 130$; $b^2 = c^2 - a^2 = 16900 - 2500 = 14400$. Equation: $\frac{x^2}{2500} - \frac{y^2}{14400} = 1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Identifying conics** • **Cross-Lesson: 9.03 The Binomial Theorem**

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

Step 1: Factor the quadratic in $\sin x$: $(2\sin x - 1)(\sin x - 1) = 0$. Step 2: Solve each factor branch:Branch 1: $2\sin x - 1 = 0 \Rightarrow \sin x = \frac{1}{2}$. On $[0, 2\pi)$, $x = \frac{\pi}{6}$ (Quadrant I) or $x = \frac{5\pi}{6}$ (Quadrant II).Branch 2: $\sin x - 1 = 0 \Rightarrow \sin x = 1$. On $[0, 2\pi)$, $x = \frac{\pi}{2}$.The complete solution set is $x \in \left\{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}\right\}$.

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16 [Challenging] • Target: Hyperbolas centered away from the origin • Cross-Lesson: 1.01 Functions and Function Notation

Solve the logarithmic equation $\log_2(x+3)+\log_2(x-3)=4$ for all real solutions x. State any extraneous roots.

Step 1: State domain restrictions: $x+3>0 \Rightarrow x>-3$ and $x-3>0 \Rightarrow x>3$, so $x>3$. Step 2: Combine logarithms: $\log_2[(x+3)(x-3)]=4 \Rightarrow \log_2(x^2-9)=4$. $x^2-9=2^4=16 \Rightarrow x^2=25 \Rightarrow x=\pm 5$. Step 3: Eliminate extraneous solutions: $x=-5$ violates $x>3$ (producing negative arguments), so it is extraneous. The unique valid solution is $x=5$.

17 [Challenging] • Target: Eccentricity and applications • Cross-Lesson: 1.02 Domain and Range

A complex number z is represented in polar form as $z=r(\cos\theta+i\sin\theta)$. Given that $z^3=8i$, and z lies in the first quadrant such that its argument θ satisfies $0<\theta<\frac{\pi}{2}$. Find the value of $r^2\cos\theta$.

We are given $z^3=8i$. In polar form, $8i=8(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2})$. Let $z=r(\cos\theta+i\sin\theta)$. Then $z^3=r^3(\cos(3\theta)+i\sin(3\theta))$. Equating the magnitudes: $r^3=8$, so $r=2$. Equating the arguments: $3\theta=\frac{\pi}{2}+2k\pi$ for integer k. Since z is in the first quadrant, $0<\theta<\frac{\pi}{2}$. For $k=0$, $3\theta=\frac{\pi}{2}$, so $\theta=\frac{\pi}{6}$. This lies in the first quadrant. For $k=1$.

18 [Challenging] • Target: Graphing a hyperbola branch • Cross-Lesson: 1.03 Average Rate of Change

Prove by mathematical induction that for all integers $n \geq 1$: $1+3+5+\ \ldots\ +(2n-1)=n^2$.

1. Base Step ($n=1$): LHS $=2(1)-1=1$, RHS $=1^2=1$. Since LHS = RHS, the base case $P(1)$ is true. 2. Inductive Hypothesis: Assume $P(k)$ is true for some integer $k \geq 1$: $1+3+5+\ \ldots\ +(2k-1)=k^2$. 3. Inductive Step ($n=k+1$): Show that $1+\ \ldots\ +(2k-1)+(2(k+1)-1)=(k+1)^2$: LHS $=[1+3+\ \ldots\ +(2k-1)]+(2k+1)$. By the inductive hypothesis, substitute k^2.

19 [Challenging] • Target: The hyperbola's defining property • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Find the infinite sum of the geometric series $12+8+\frac{16}{3}+\frac{32}{9}+\ \ldots$. Verify the condition for convergence.

Step 1: Determine first term and common ratio: First term $a_1=12$. Common ratio $r=\frac{8}{12}=\frac{2}{3}$. Step 2: Check convergence condition: Since $|r|=\left|\frac{2}{3}\right|=\frac{2}{3}<1$, the infinite series converges. Step 3: Compute infinite sum: $S_\infty=\frac{a_1}{1-r}=\frac{12}{1-2/3}=\frac{12}{1/3}=36$.

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20 [Challenging] • Target: Comparing all four conic sections • Cross-Lesson: 2.02 Radian Measure and Arc Length

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

For a rational function to have a removable discontinuity at $x = c$, the numerator must have a factor of $(x - c)$ that cancels with a factor in the denominator. Given a removable discontinuity at $x = 2$, the numerator $x^2 - ax + 6$ must have $x = 2$ as a root. Thus, $2^2 - a(2) + 6 = 0$, which simplifies to $4 - 2a + 6 = 0$, so $10 - 2a = 0$, and $a = 5$. The numerator can be factored as $(x - 2)(x - 3)$ since $2 \times 3 = 6$ and $2 + 3 = 5$, so $a = 5$. For the function to have a vertical asymptote at $x = 3$, the.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Standard Equations of Hyperbolas

Find the vertices and orientation of $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. **Horizontal, vertices at** $(\pm 4, 0)$ ✓

b. Vertical, vertices at $(0, \pm 3)$

c. Horizontal, vertices at $(\pm 5, 0)$

d. Vertical, vertices at $(0, \pm 4)$

Positive term is x^2, so the transverse axis is horizontal. $a^2 = 16 \Rightarrow a = 4$. Vertices are $(\pm 4, 0)$.

22 [Foundational] • Target: Standard Equations of Hyperbolas

Find the vertices and orientation of $\frac{y^2}{25} - \frac{x^2}{144} = 1$.

a. **Vertical, vertices at** $(0, \pm 5)$ ✓

b. Horizontal, vertices at $(\pm 12, 0)$

c. Vertical, vertices at $(0, \pm 13)$

d. Horizontal, vertices at $(\pm 5, 0)$

Positive term is y^2, so the transverse axis is vertical. $a^2 = 25 \Rightarrow a = 5$. Vertices are $(0, \pm 5)$.

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23 [Foundational] • Target: Standard Equations of Hyperbolas

Find the length of the transverse axis and conjugate axis for $\frac{(x-1)^2}{36} - \frac{(y+2)^2}{64} = 1$.

a. **Transverse** $= 12$**, Conjugate** $= 16$ ✓

b. Transverse $= 6$, Conjugate $= 8$

c. Transverse $= 16$, Conjugate $= 12$

d. Transverse $= 72$, Conjugate $= 128$

$a^2 = 36 \Rightarrow a = 6 \Rightarrow 2a = 12$. $b^2 = 64 \Rightarrow b = 8 \Rightarrow 2b = 16$.

24 [Foundational] • Target: Asymptotes of Hyperbolas

State the equations of the asymptotes for the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

a. $y = \pm\frac{b}{a}x$ ✓

b. $y = \pm\frac{a}{b}x$

c. $y = \pm\frac{b^2}{a^2}x$

d. $y = \pm x$

For a horizontal hyperbola centered at the origin, the asymptotes pass through the center with slopes $\pm b/a$: $y = \pm\frac{b}{a}x$.

25 [Foundational] • Target: Asymptotes of Hyperbolas

Find the asymptotes of $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

a. $y = \pm\frac{4}{3}x$ ✓

b. $y = \pm\frac{3}{4}x$

c. $y = \pm\frac{16}{9}x$

d. $y = \pm\frac{5}{3}x$

$a = 3, b = 4$. Asymptotes: $y = \pm\frac{b}{a}x = \pm\frac{4}{3}x$.

26 [Foundational] • Target: Asymptotes of Hyperbolas

Find the asymptotes of the vertical hyperbola $\frac{y^2}{4} - \frac{x^2}{9} = 1$.

a. $y = \pm\frac{2}{3}x$ ✓

b. $y = \pm\frac{3}{2}x$

c. $y = \pm\frac{4}{9}x$

d. $y = \pm 2x$

For a vertical hyperbola centered at $(0, 0)$, a^2 is under y^2 and b^2 is under x^2. Asymptotes are $y = \pm\frac{a}{b}x = \pm\frac{2}{3}x$.

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27 [Foundational] • Target: Foci and Eccentricity of Hyperbolas

What is the relationship between a, b, and focal distance c for a hyperbola?

a. $c^2 = a^2 + b^2$ ✓

b. $c^2 = a^2 - b^2$

c. $a^2 = b^2 + c^2$

d. $c = a + b$

In a hyperbola, the foci lie farther from the center than the vertices, so $c^2 = a^2 + b^2$.

28 [Foundational] • Target: Foci and Eccentricity of Hyperbolas

Find the foci of $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. $(\pm 5, 0)$ ✓

b. $(0, \pm 5)$

c. $(\pm\sqrt{7}, 0)$

d. $(\pm 25, 0)$

$c^2 = 16 + 9 = 25 \Rightarrow c = 5$. Horizontal transverse axis: foci are $(\pm 5, 0)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Foci and Eccentricity of Hyperbolas

Find the eccentricity $e = c/a$ of $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. $5/4 = 1.25$ ✓

b. $4/5 = 0.8$

c. $3/4 = 0.75$

d. $\sqrt{7}/4$

$c = 5, a = 4$. Eccentricity $e = c/a = 5/4 = 1.25 > 1$.

30 [Intermediate] • Target: Identifying Conics from General Second-Degree Equations

For $Ax^2 + Cy^2 + Dx + Ey + F = 0$ ($A, C \neq 0$), which condition identifies a hyperbola?

a. A **and** C **have opposite signs (**$AC < 0$**)** ✓

b. A and C have the same sign and $A = C$

c. A and C have the same sign and $A \neq C$

d. $A = 0$ or $C = 0$

If $AC < 0$, the quadratic terms have opposite signs, representing a hyperbola.

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31 [Intermediate] • Target: Identifying Conics from General Second-Degree Equations

Identify the conic: $9x^2 - 4y^2 - 36x + 8y - 4 = 0$.

a. **Hyperbola** ✓

b. Ellipse

c. Parabola

d. Circle

$A = 9 > 0$ and $C = -4 < 0$. The signs are opposite, so this equation represents a hyperbola.

32 [Intermediate] • Target: Identifying Conics from General Second-Degree Equations

Convert $9x^2 - 4y^2 - 36x + 8y - 4 = 0$ to standard form.

a. $\frac{(x-2)^2}{4} - \frac{(y-1)^2}{9} = 1$ ✓

b. $\frac{(x-2)^2}{9} - \frac{(y-1)^2}{4} = 1$

c. $\frac{(y-1)^2}{9} - \frac{(x-2)^2}{4} = 1$

d. $(x-2)^2 - (y-1)^2 = 36$

$9(x^2 - 4x + 4) - 4(y^2 - 2y + 1) = 4 + 36 - 4 = 36$. Dividing by 36: $\frac{(x-2)^2}{4} - \frac{(y-1)^2}{9} = 1$.

33 [Intermediate] • Target: Identifying Conics from General Second-Degree Equations

Find the asymptotes of the hyperbola $\frac{x^2}{25} - \frac{y^2}{9} = 1$.

a. $y = \pm\frac{5}{3}x$

b. $y = \pm\frac{3}{5}x$ ✓

c. $y = \pm 15x$

d. $y = \pm\frac{8}{5}x$

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x = \pm\frac{3}{5}x$.

34 [Intermediate] • Target: Standard form

Find the asymptotes of the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$.

a. $y = \pm\frac{5}{4}x$

b. $y = \pm 20x$

c. $y = \pm\frac{9}{5}x$

d. $y = \pm\frac{4}{5}x$ ✓

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x = \pm\frac{4}{5}x$.

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35 [Intermediate] • Target: Identifying conics

Find the asymptotes of the hyperbola $\frac{x^2}{36} - \frac{y^2}{9} = 1$.

a. $y = \pm\frac{1}{2}x$ ✓

b. $y = \pm 18x$

c. $y = \pm 2x$

d. $y = \pm\frac{3}{2}x$

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x = \pm\frac{1}{2}x$.

36 [Intermediate] • Target: Hyperbolas centered away from the origin

Find the asymptotes of the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$.

a. $y = \pm 6x$

b. $y = \pm\frac{2}{3}x$ ✓

c. $y = \pm\frac{3}{2}x$

d. $y = \pm\frac{5}{3}x$

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x = \pm\frac{2}{3}x$.

37 [Intermediate] • Target: Eccentricity and applications

Find the asymptotes of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. $y = \pm\frac{4}{3}x$

b. $y = \pm 12x$

c. $y = \pm\frac{7}{4}x$

d. $y = \pm\frac{3}{4}x$ ✓

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x = \pm\frac{3}{4}x$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Graphing a hyperbola branch • Cross-Lesson: 1.03 Average Rate of Change

Find the asymptotes of the hyperbola $\frac{x^2}{4} - \frac{y^2}{1} = 1$.

a. $y = \pm\frac{1}{2}x$ ✓

b. $y = \pm 2x$

c. $y = \pm\frac{3}{2}x$

d. $y = \pm -\frac{1}{2}x$

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x = \pm\frac{1}{2}x$.

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39 [Challenging] • Target: The hyperbola's defining property • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Find the asymptotes of the hyperbola $\frac{x^2}{9} - \frac{y^2}{1} = 1$.

a. $y = \pm 3x$

b. $y = \pm\frac{1}{3}x$ ✓

c. $y = \pm\frac{4}{3}x$

d. $y = \pm -\frac{2}{3}x$

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x = \pm\frac{1}{3}x$.

40 [Challenging] • Target: Comparing all four conic sections • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the asymptotes of the hyperbola $\frac{x^2}{25} - \frac{y^2}{1} = 1$.

a. $y = \pm -\frac{4}{5}x$

b. $y = \pm\frac{6}{5}x$

c. $y = \pm 5x$

d. $y = \pm\frac{1}{5}x$ ✓

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x = \pm\frac{1}{5}x$.

41 [Challenging] • Target: Finding the equation from asymptotes and a point • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Find the asymptotes of the hyperbola $\frac{x^2}{36} - \frac{y^2}{16} = 1$.

a. $y = \pm\frac{2}{3}x$ ✓

b. $y = \pm 24x$

c. $y = \pm\frac{5}{3}x$

d. $y = \pm\frac{3}{2}x$

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x = \pm\frac{2}{3}x$.

42 [Challenging] • Target: Applications: LORAN navigation • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Find the asymptotes of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

a. $y = \pm 12x$

b. $y = \pm\frac{7}{3}x$

c. $y = \pm\frac{3}{4}x$

d. $y = \pm\frac{4}{3}x$ ✓

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x = \pm\frac{4}{3}x$.

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43 [Challenging] • Target: Standard Equations of Hyperbolas • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Find the asymptotes of the hyperbola $\frac{x^2}{16} - \frac{y^2}{4} = 1$.

a. $y = \pm\frac{1}{2}x$ ✓

b. $y = \pm 8x$

c. $y = \pm\frac{3}{2}x$

d. $y = \pm 2x$

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x = \pm\frac{1}{2}x$.

44 [Challenging] • Target: Asymptotes of Hyperbolas • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Find the asymptotes of the hyperbola $\frac{x^2}{25} - \frac{y^2}{4} = 1$.

a. $y = \pm 10x$

b. $y = \pm\frac{5}{2}x$

c. $y = \pm\frac{7}{5}x$

d. $y = \pm\frac{2}{5}x$ ✓

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x = \pm\frac{2}{5}x$.

45 [Challenging] • Target: Foci and Eccentricity of Hyperbolas • Cross-Lesson: 3.03 Sum and Difference Formulas

Find the asymptotes of the hyperbola $\frac{x^2}{16} - \frac{y^2}{1} = 1$.

a. $y = \pm\frac{1}{4}x$ ✓

b. $y = \pm\frac{5}{4}x$

c. $y = \pm -\frac{3}{4}x$

d. $y = \pm 4x$

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x = \pm\frac{1}{4}x$.

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